Automatic Fall Detection using Deep Neural Networks with Aggregated Residual Transformation

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2022 37th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC) | 978-1-6654-8559-3/22/$31.00 ©2022 IEEE | DOI: 10.1109/ITC-CSCC55581.2022.9895054

***Abstract*—The advancement of inertial sensor technology and the growing prevalence of intelligent wearables (including smartwatches, smart bands, and other intelligent devices) have aided in applying science and technology to automated Fall Detection Systems (FDSs). Over the past decade, wearable-based FDSs has garnered considerable scientific attention. In this context, machine learning (ML) techniques have demonstrated remarkable efficacy in differentiating between falls and typical motions or Activities of Daily Living (ADLs) using the data recorded by wearable inertial sensors. Unfortunately, in most research, the effectiveness of machine learning classifiers was restricted by feature extraction and selection processes that relied on human-crafted decisions. A deep neural network was established to enhance the capacity of fall detection, which combines convolutional layers and aggregated residual transformation in this study. The TensorFlow framework trained the proposed model to identify and categorize 15 special events: three distinct falls and twelve ADLs. On a publicly available benchmark dataset (UMAFall), the proposed model was analyzed and compared to existing baseline models. The findings indicate that the proposed model is superior to other models, with an accuracy of 96.72%.**

***Keywords*—fall detection, wearable sensors, deep learning, aggregated residual transformation**

I. Introduction

Aging is a global issue that affects life expectancy. According to the World Health Organization (WHO), the aged population accounts for 20% of the worldwide population [1]. According to another projection, the number of individuals over 65 will rise to 1.5 billion by 2050. By and large, aging impairs physical, cognitive, and sensory abilities [2]. As a result, an older adult may have trouble with routine activities such as walking, jogging, eating, and dressing. Falling is an important issue for the elderly since it may significantly lower the expected lifespan.

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A fall is an unanticipated occurrence that forces someone to remain on a lower level. As a consequence, it commonly causes fatal wounds [3]. Additionally, psychological discomfort is considered a consequence of falls. People can suffer anxiety, depression, activity restriction, and fear of falling. Fear of falling is the greatest physiological issue in the elderly, limiting their Activities of Daily Living (ADL) [4], [5].

Recent technological advancements have paved the way for the creation of Assisted-Living Systems (ALSs) [6]. They can aid the elderly and maintain a secure environment by alleviating caretakers of their effort. Nonetheless, ALSs introduce other obstacles, such as confidentiality and acceptability considerations, that must be considered.

Fall Detection Systems (FDSs), one type of ALSs, aim to identify falls and notify caregivers to react as rapidly as possible. Fall recognition is, unfortunately, computationally intensive. Falls are abrupt transitions from an upright to a supine or virtually extended position accompanied by uncontrolled movement [7]. Increasing acceleration occurs with falls. Additionally, falls may occur in an endless variety of ways. They can occur at any time and in any place. Their initial and ultimate body positions and their direction (forward, backward) might vary. Therefore, FDSs must encompass the whole dwelling area. Their reliability must be considered while preventing false alarms and protecting the elderly's privacy.

Over the past decade, machine learning (ML) algorithms have demonstrated remarkable efficacy in differentiating between falls and typical motions or ADLs using the data recorded by wearable inertial sensors. Four well-known and commonly utilized ML techniques in fall detection systems are Random Forest, Support Vector Machine, Multi-layer Perceptron, and K-nearest Neighbors. Nonetheless, the effectiveness of these ML algorithms was constrained by hand-crafted feature extraction.

Deep Learning (DL) has been widely used in most industries

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worldwide recently [8]–[10]. DL methods have been used for fall detection [11], [12] with more effectiveness than threshold-based strategies in the last few years. Multiple layers compose DL models, with each layer deriving or transforming a feature from the input data. The last tiers of models often consist of artificial neurons. For instance, the data may take the form of a vision-recorded picture, raw data from an accelerometer, or a gyroscope. Numerous DL approaches are utilized to recognize fall occurrences. Some solutions are equipped only via a single DL approach, while others are constructed through many strategies to achieve greater recognition accuracy [13], [14]. The most often utilized DL models for fall detection are convolutional neural networks (CNNs) and recurrent neural networks (RNNs) [15]–[17].

So, we developed a deep neural network in this study using convolutional layers and aggregated residual processing. The proposed model was trained using the TensorFlow platform to identify and characterize ten particular incidents: three distinct falls and twelve ADLs. The proposed model was analyzed and compared against various baseline deep learning algorithms (UMAFall).

II. The Fall Detection Framework

The sensor-based framework of fall detection, which is used in this work, composes of four main processes: data acquisition, data pre-processing, data generation, model training and interpretation as shown in Fig. 1.

A. UMAFall Dataset

UMAFall [18] is a publicly available dataset that gathers data on three kinds of falls and twelve ADLs using a smartphone located in the suitable thigh area, including four sensors worn on the waist, ankle, chest, and right wrist. Seventeen volunteers (seven females and ten males) performed a minimum of three investigations of each action in a household setting. Fall-backward, fall-forwards, and fall-lateral are the three distinct types of falls. Moreover, the 12 ADLs are applauding, body bending (squatting), climbing stairs up, ascending stairs down, lifting hands, jumping, gentle jogging, laying down to rest, speaking on the phone, opening a door, seating, and walking. Table I summarizes the main features of the 17 users who engaged in the investigations.

TABLE I. Participant details of UMAFall dataset

Sex	Age (years)	Height (cm.)	Weight (kg.)
7 females and 10 males	18-55	155-195	50-93

B. Data Pre-processing

The sensor readings data were processed during the preliminary data processing by having the following operations carried out on them. Firstly, the noise decreased using a median filter and a low-pass Butterworth filter of the third order with a cutoff frequency of 20 Hz. Then, the normalizing of the data through the application of the Min-Max method. The data from the preprocessed sensors were then divided employing fixed-width sliding windows with an overlap of one second comprised of fifty percent of the total time.

C. Training Models for Deep Learning

We established a multi-branch aggregation approach in this study influenced by ResNeXt [19]. In this model, the different sized kernel feature maps are contributed rather than concatenating the various sized kernel feature maps like that in InceptionNet [20]. This drastically reduces the model's parameters, making such interconnections appropriate for edge and low latency operations. Fig. 2 illustrates the FallNeXt in further information.

The offered model, named FallNeXt model, consists of four modules with convolutional kernels of varying sizes. Each MultiKernel (MK) module has three distinct kernel sizes: 1 $\times$ 3, 1 $\times$ 5, and 1 $\times$ 7. Additionally, 1 $\times$ 1 convolutions are utilized before employing these kernels to decrease the mode's sophistication of the system and parameter amount. Fig. 3 illustrates the MultiKernel in perspective.

III. Experiments and Results

The purpose of this section is to describe the experimental design and results operated to evaluate the proposed FallNeXt model for fall detection in a range of scenarios.

A. Experiment Setting

Each investigation in this study was achieved using a Google Colab Pro+ system utilizing a Tesla V100. In addition to Python, TensorFlow, Keras, Scikit-Learn, Numpy, and Pandas, the Python programming language is also developed utilizing these libraries. We conduct tests to evaluate the identification abilities of the proposed FallNeXt model and compare it to standard DL algorithms and ResNet-based models. There are 18, 34, 50, and 101 CNN layers inside ResNet. We separated these investigations into three scenarios to examine fall behavior across ages: Scenario-I used only ADL data, Scenario-II used just FALL data, and Scenario-III utilized both ADL and FALL data.

B. Experimental Results

Filtering the UMAFall dataset was accomplished based on ADL activity sensor data. A five-fold cross-validation process was utilized to obtain these results. Five baseline deep learning models and four ResNet-based models were evaluated in this study via a series of tests. Various measures, including accuracy, loss, and F1-score, are suggested for FallNeXt, as indicated in Table II.

As a consequence of the findings, it was determined that the recommended FallNeXt model attained the maximum accuracy and F1-score with the fewest parameters. Tables III and IV confirm that the FallNeXt model surpasses other standard DL models and ResNet-based models under various circumstances.

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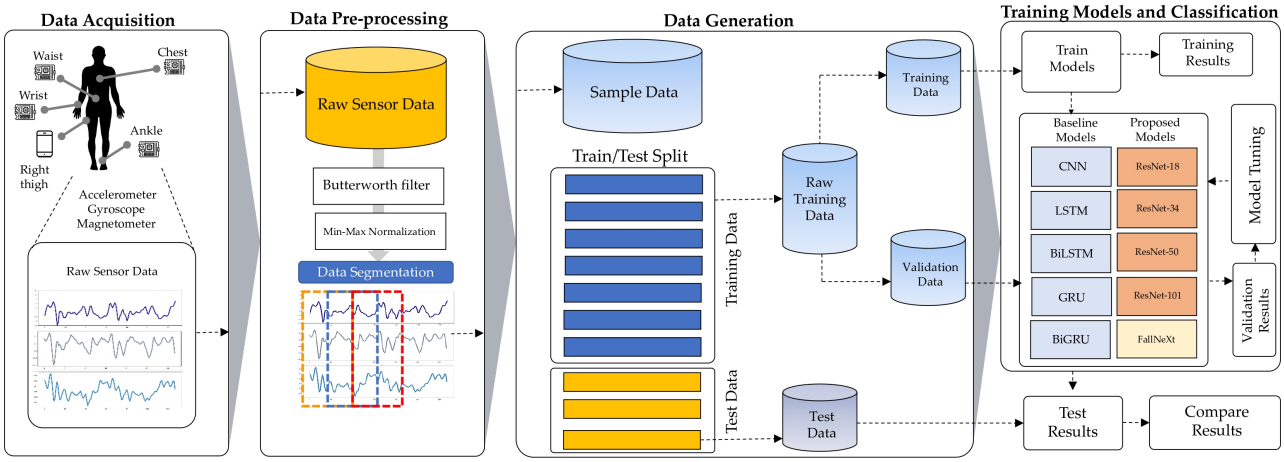


Fig. 1. The fall detection framework based on wearable sensors used in this work

TABLE II. Effectiveness model evaluation using ADL data from the UMAFall dataset

Model	Parameter	Identification effectiveness		
		Accuracy	Loss	F1-score
Baseline DL models				
CNN	33,804	87.03(±1.662)	0.71(±0.061)	85.15(±1.576)
LSTM	104,076	92.64(±0.526)	0.28(±0.020)	91.83(±0.661)
BiLSTM	206,476	93.72(±0.330)	0.27(±0.024)	92.95(±0.609)
GRU	82,956	91.89(±0.799)	0.31(±0.030)	91.19(±0.561)
BiGRU	164,236	93.43(±0.647)	0.25(±0.026)	92.76(±0.382)
ResNet-based DL models				
ResNet-18	4,250,572	93.65(±0.992)	0.32(±0.047)	92.23(±1.451)
ResNet-34	8,206,476	90.62(±1.113)	0.50(±0.073)	89.12(±1.315)
ResNet-50	30,723,276	91.33(±0.907)	0.42(±0.033)	90.33(±0.438)
ResNet-101	60,573,132	90.59(±0.230)	0.49(±0.042)	88.34(±0.648)
Proposed models				
FallNeXt	27,622	95.71(±1.547)	0.24(±0.047)	95.14(±1.867)

TABLE III. Effectiveness model evaluation using FALL data from the UMAFall dataset

Model	Parameter	Identification effectiveness		
		Accuracy(%)	Loss	F1-score(%)
Baseline DL models				
CNN	33,507	97.36(±0.473)	0.27(±0.092)	97.32(±0.482)
LSTM	102,915	97.14(±0.702)	0.13(±0.039)	97.10(±0.695)
BiLSTM	205,315	97.06(±0.717)	0.13(±0.033)	97.03(±0.712)
GRU	81,795	97.73(±0.483)	0.10(±0.023)	97.72(±0.475)
BiGRU	163,075	97.83(±0.512)	0.09(±0.032)	97.82(±0.532)
ResNet-based DL models				
ResNet-18	4,250,275	97.54(±0.601)	0.13(±0.033)	97.51(±0.581)
ResNet-34	8,206,179	96.48(±1.054)	0.22(±0.074)	96.43(±1.058)
ResNet-50	30,722,979	96.77(±0.891)	0.18(±0.043)	96.71(±0.905)
ResNet-101	60,572,835	95.27(±0.994)	0.20(±0.053)	95.71(±0.905)
Proposed model				
FallNeXt	27,037	99.18(±0.446)	0.04(±0.024)	99.19(±0.442)

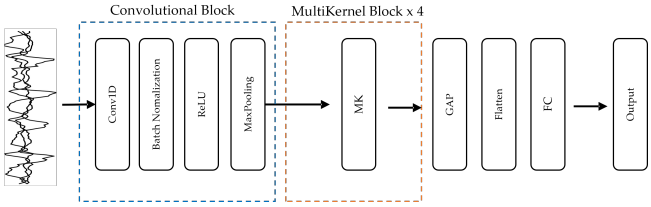


Fig. 2. The proposed FallNeXt model

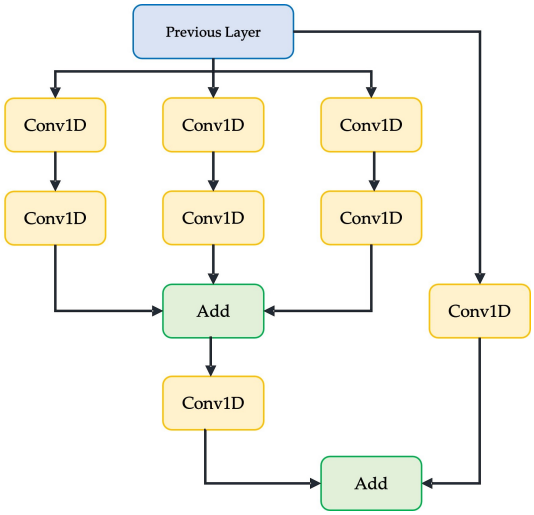


Fig. 3. Structure of a MultiKernel module

These experimental findings indicate that the constructed DL model with aggregated residual transformation called FallNeXt model could significantly enhance the fall detection experience in various testing conditions.

IV. Conclusion and Future Study

We proposed a DL model based on aggregated residual transformations to address the fall detection issue by effectively utilizing wearable sensors in this work. FallNeXt is

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TABLE IV. Effectiveness model evaluation using all data from the UMAFall dataset

Model	Parameter	Identification effectiveness		
		Accuracy(%)	Loss	F1-score(%)
Baseline DL models				
CNN	33,903	84.71(±1.125)	0.77(±0.076)	82.50(±1.324)
LSTM	104,463	92.25(±0.213)	0.29(±0.010)	91.50(±0.664)
BiLSTM	206,863	93.48(±0.454)	0.27(±0.011)	92.76(±0.616)
GRU	83,343	91.38(±0.604)	0.31(±0.021)	90.02(±0.698)
BiGRU	164,623	93.07(±0.451)	0.27(±0.027)	92.54(±0.729)
ResNet-based DL models				
ResNet-18	4,250,671	94.14(±0.621)	0.25(±0.028)	93.49(±0.610)
ResNet-34	8,206,575	92.52(±0.733)	0.32(±0.034)	90.54(±0.917)
ResNet-50	30,723,375	93.15(±0.580)	0.30(±0.069)	91.99(±1.016)
ResNet-101	60,573,231	87.96(±1.225)	0.43(±0.076)	85.59(±1.125)
Proposed models				
FallNeXt	27,817	96.72(±0.449)	0.19(±0.023)	96.51(±0.586)

the name of the proposed deep learning model, which uses some of the benefits of multi-kernel modules. We investigated the recommended FallNeXt model using a publicly available benchmark dataset for wearable sensors, UMAFall. The FallNeXt model surpasses other activity-based models with the maximum accuracy of 96.72% when ADL and FALL activities are included. The FallNeXt model surpasses other activity-based models with the maximum accuracy of 96.72% when ADL and FALL activities are included.

Future studies should evaluate the proposed DL models using a more comprehensive sample size of participants with varying fall styles. Developing more complicated and lightweight DL networks and innovative time-frequency-based data representations might improve additional performance improvements.

ACKNOWLEDGMENT

This research project was supported by the Thailand Science Research and Innovation fund; the University of Phayao (Grant No. FF65-RIM041); National Science, Research and Innovation Fund (NSRF); and King Mongkut's University of Technology North Bangkok with Contract no. KMUTNB-FF-66-07.

The authors also gratefully acknowledge the support provided by Thammasat University Research fund under the TSRI, Contract No. TUFF19/2564 and TUFF24/2565, for the project of "AI Ready City Networking in RUN", based on the RUN Digital Cluster collaboration scheme.

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